PART V · 25 MINUTES

Design patterns for never-ending NLP systems

How do we turn update targets into a loop that improves the system without destabilizing it?

Patterns for self-improving language agents

1 Knowledge-grounded retrieval and memory updates
2 Substrate choice: modular vs. end-to-end
3 Consolidation and selective forgetting
4 Coupled views and tasks without circularity
5 Provenance and auditability
6 The governed update lifecycle and validation gates
7 Staging, shadow mode and quarantine
8 Version bundles, rollback and repair
9 Bounded autonomy and human curation
10 Reflective improvement grounded in evidence

1-Retrieve structure and evidence, not only passages

For every output:

- Which evidence supports the answer?
- Which claims are uncertain or stale?
- What new candidate knowledge emerged?

StackFeed: a structured textual actor, critic that edits a knowledge base using feedback.

Kirtania et al., 2025

MemGov: detects must-link and cannot-link constraints among retrieved memories. Christou et al., AACL2026

see also LLM Wiki in Part VI

1-Every memory write should be a controlled update

Write path:
Mem0 (Chhikara et al., 2025); A-MEM links and evolves notes (Xu et al., 2025)

Read path: Retrieve by relevance **and learned usefulness**: the most similar memory is not always the one that prevents the failure.
MemRL (Zhang et al., 2026)

2-Modular versus end-to-end: where should learning live?

Parametric

continual fine-tuning · adapters / LoRA · RL

+ broad generalization, cheaper inference

– effects are distributed; hard to inspect, scope or remove a single influence

Non-parametric

KB · memory · prompts and context · skill library · evaluators

+ local, inspectable, versionable, reversible

– interference, context crowding, unexpected interactions

2-Prompts, contexts and harnesses as learned artifacts

GEPA: Reflects on trajectories to propose and test prompt changes. Agrawal et al., 2025

ACE: Context as an evolving playbook: generate, reflect, curate. Zhang et al., 2026

Test-time learning: Adapting an LLM during deployment on the inputs it sees. Hu et al., 2025

RPT: Reflective Prompt Tuning through Language Model Function-Calling, Bayat et al., 2026

An optimizer that repeatedly selects prompts against one dev set **adapts to that set's peculiarities**.

3-Consolidation moves learning across substrates and can lose it

COMPRESSION HAZARD

“Do not inherit training datasets without explicit support in derived-model documentation”

Can be summarized as:

“Do not inherit facts.” — shorter, broader, harmful

Test every summary on positive and negative cases. Keep links to evidence. Version the datasets.

3-Operations that are often confused

Operation	What happens
Revision	A belief or rule changes in response to evidence
Forgetting	Previously available information or capability is lost
Expiration	An item is withdrawn under a validity or retention policy
Deletion	A representation is removed

4-Multiple views can be complementary without being independent

View — a representation or channel: local sentence, document structure, typed KB context.

Source — the origin of an observation. One source can feed several views.

Each view contributes **within its evidential scope**

Validate a view's labels with sources or procedures that **do not repeat its predictions**.

Co-training assumes conditions on the views (Blum and Mitchell, 1998).

5-Every learned change should be able to explain itself

For each learned claim or rule

Who or what produced it?

At what time?

By which method?

From which source?

With what confidence?

Approved by whom?

And: what depends on it now?

6-stages between an observation and active state

Observation: record the event and context

Proposal: a candidate change and its intended benefit

Assessment: technical evidence about the candidate

Authorization: does policy permit activation?

Commitment: a versioned active state

Monitoring: consequences not visible before; may open a new cycle

Any stage may end in rejection, revision or staging, not only in commitment.

7-Limit exposure before you trust a change

Shadow mode: Run the candidate on representative inputs and compare outputs — without writing accepted state.

Quarantine ≠ deletion: A questionable record stays visible to auditors but is excluded from supervision and ordinary retrieval.

Staged promotion: Start with one domain or relation family. Expanding needs evidence about the newly exposed conditions.

8-Version compatible bundles, not individual files

- **Why bundles**: Prompt p4 assumes schema s2 and index r7. Reverting p4 alone can produce an invalid configuration if a later schema removed a relation name.
- **What to record:** before/after states · change sets · provenance · affected records and inferences · outputs already delivered
- **Rollback:** Replace the harmful state with an earlier compatible bundle. Simple when little has happened since.
- **Repair:** Trace descendants; keep those with independent support; quarantine the rest; preserve unrelated progress.

9-Autonomy is decided stage by stage, not all at once

Stage	Who decides	Why
Detect an error pattern	Agent	Cheap, reversible
Propose a new task	Agent	A proposal changes nothing
Collect examples, train a candidate	Agent, in a sandbox	No effect on active state
Recommend promotion	Agent + evaluator	Evidence, not permission
Activate in production	Policy or human	Affects users
Change the acceptance rule	Never the agent alone	Avoid circular validation

Bounded autonomy is not guaranteed safety.

9-Spend scarce human attention where it changes outcomes

Prioritize for review

- high-impact updates, even when confident
- conflicting evidence
- schema and policy changes
- proposed new learning tasks
- a random sample, to find blind spots

Reviewer decisions are themselves trusted signals and review effort belongs in the cost of learning.

10-Reflection proposes; evidence decides

Reflection can happen:

- **Within an episode:** Iterative self-feedback and revision of outputs. Self-Refine · Madaan et al., 2023
- **Across trials:** Verbal lessons stored and reused on later attempts. Reflexion · Shinn et al., 2023

Must be Grounded in supervision: Instance-level critiques plus reusable semantic guidance from labeled examples. Hassell et al., 2026

KEY MESSAGE · PART IV

Design the update path as carefully as the inference path: typed state, explicit lineage, gated promotion and reversible change.

Good governance makes decisions explicit and repair possible. It does not certify correctness by construction.

Next · Part V — evaluation, governance and reliability under change.